

CONVERSATIONAL CLUSTERING

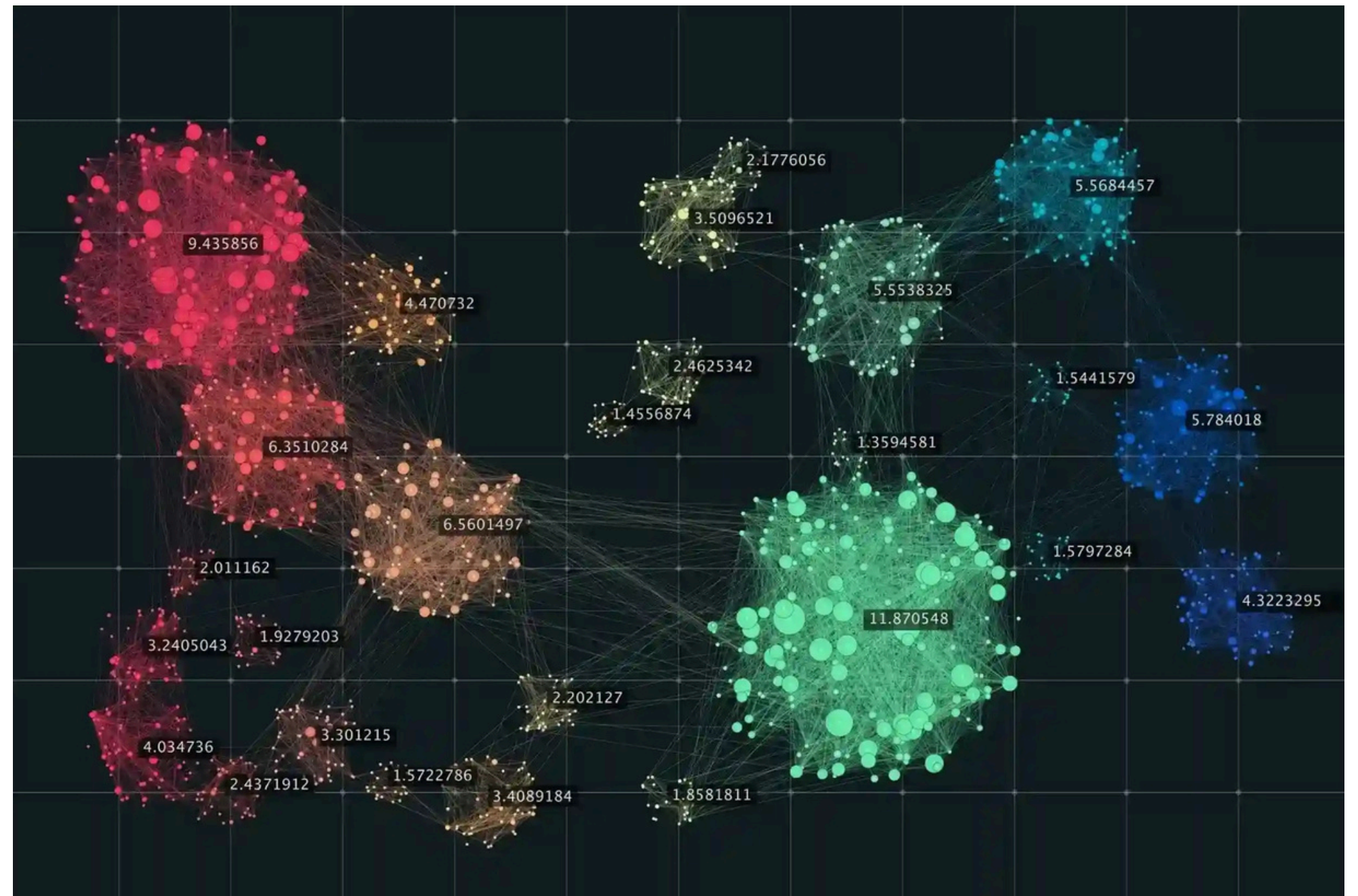
PROBLEM STATEMENT

TECHNICAL KNOWLEDGE

Users often lack technical knowledge about clustering techniques

FINETUNING

Translating high-level intents into technical choices places the cognitive burden in the wrong place



RESEARCH QUESTIONS

QUESTION 1

Which question type is the best to ask (yes/no, multiple choice, open-ended), in terms conversation turns, tokens and answers length?

QUESTION 2

How faithful is the proposed result, given the preferences of the user?

STATE OF THE ART

TAKEN FROM PRIOR WORK

- Instruction-conditioned embeddings
- Conversational preference elicitation
- Clustering algorithms

HAD TO FIGURE OUT

- Elicitation without model fine tuning
- Conversation evaluation metrics
- Subjective clustering metrics (no ground truth)

RELEVANT PAPERS

- Asking Clarifying Questions for Preference Elicitation With Large Language Models
- Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models
- Large Language Models Enable Few-Shot Clustering

ARCHITECTURE

1. CONVERSATION

The users explains its clustering preferences to the LLM.
If needed, clarifying questions are asked by the model, to better frame the user's intentions

2. EMBEDDINGS GENERATION

Use Qwen3-Embedding to encode all the documents in the dataset.
The user's preferences are also taken into account.

3. CLUSTERING

A clustering algorithm is run on the embedded documents.
Strategies such as summary-as-centroids or keyphrase generation can be used to improve the results.

DATASET

STACKOVERFLOW

Around 20k short texts (question titles).

Clustering can be based on programming language, architectural vs debugging questions, etc.

EXPERIMENTS & EVALUATION

EXPERIMENTS

- 108 total experiments (36 per question type)
- 3 different personas
- 6 distinct goals

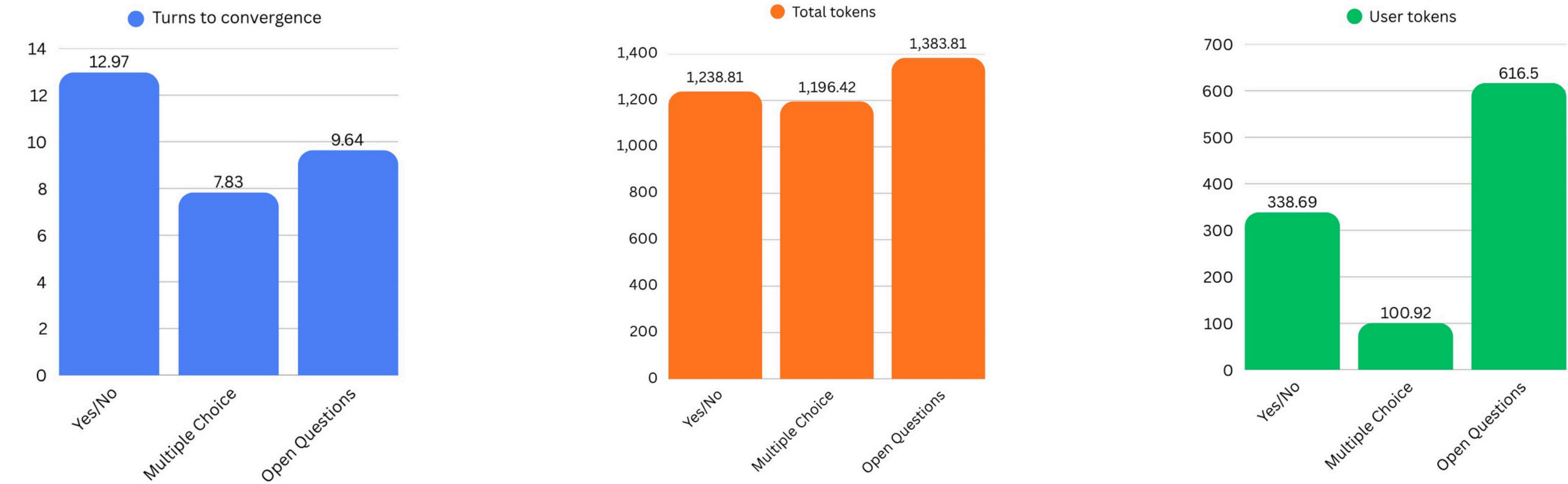
QUESTION 1

- Turns to convergence
- Total tokens
- Answer tokens

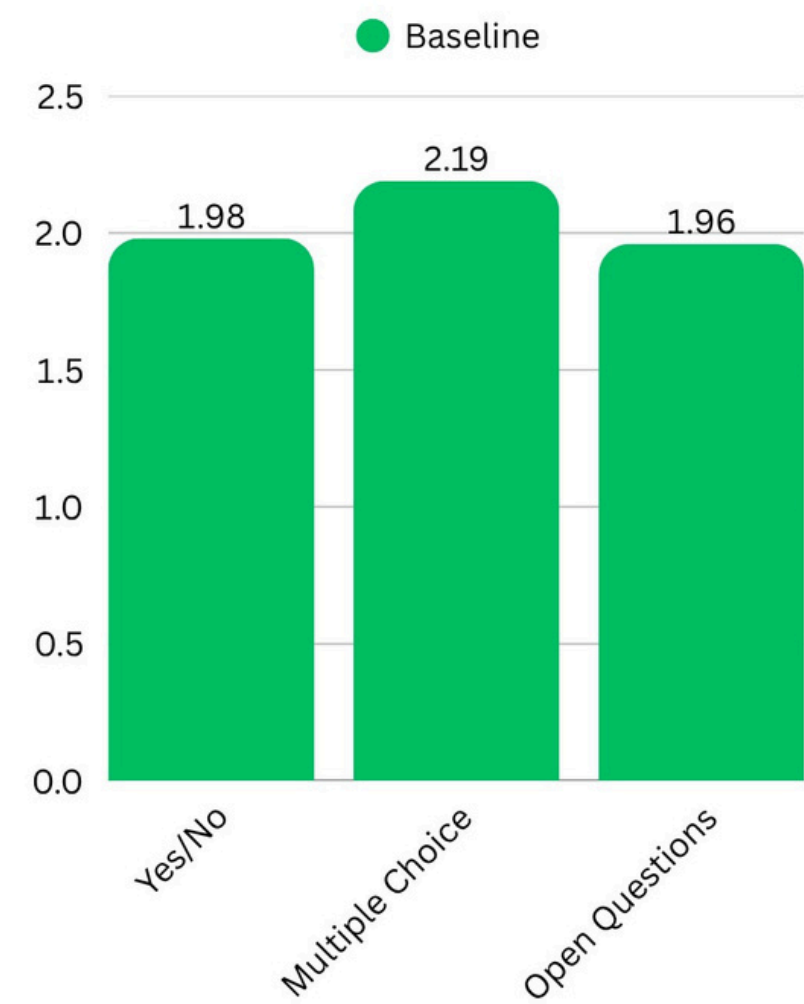
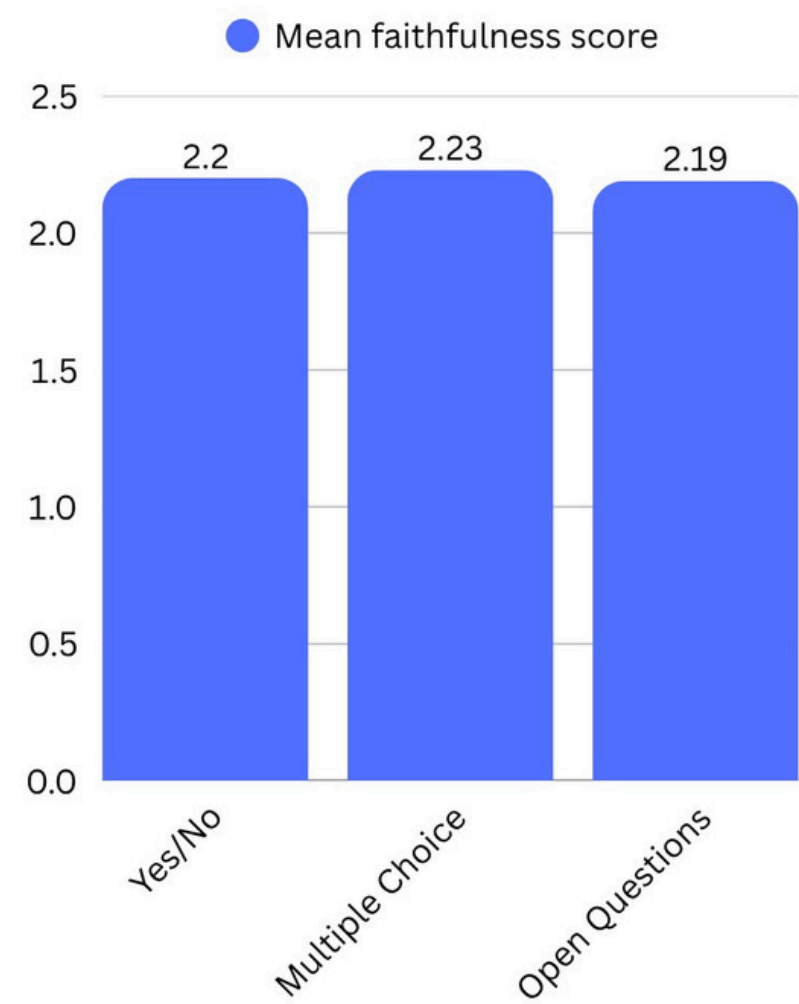
QUESTION 2

- LLM-as-judge
- Faithfulness score:
 - Coherence
 - Alignment
 - Separation

RESULTS – Q1



RESULTS – Q2



LIMITATIONS

DATASET

Extremely short texts, hard to correctly embed

SIMULATED USERS

Evaluated on synthetic profiles only rather than authentic human interaction data

USER PREFERENCE TO EMBEDDING INSTRUCT

Hard to evaluate how the results are conditioned by the preference summarization and by the encoder model instruction

LIMITED HUMAN IN THE LOOP VALIDATION

Small human annotated samples to compute Cohen's K